

Neural networks:

How a neuron integrates its inputs:

$$C \frac{dV}{dt} = -G_L (V - E_L) + \sum_{i=1}^N W_i \sum_{j=1}^{n_i} K_s(t - t_j^{(i)})$$

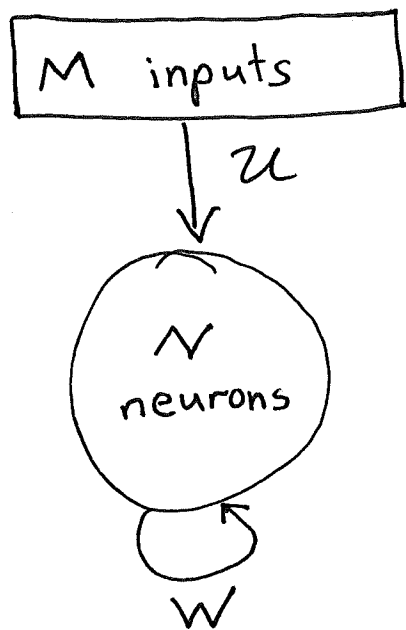
Consider a population of N neurons connected to each other through a synaptic weight matrix, W :

W_{ij} : synaptic weight $j \rightarrow i$ (recurrent);

and driven by M input neurons through a matrix, U :

U_{ij} : synaptic weight $j \rightarrow i$ (input)

Diagram:



Coupled equations for neural network, $i=1, \dots, N$

$$C \frac{dV_i}{dt} = -G_L (V_i - E_L) + \sum_{j=1}^N W_{ij} \sum_{k=1}^{n_j} K_s(t - t_k^{(j)}) + \sum_{j=1}^M U_{ij} \sum_{k=1}^{m_j} K_s(t - \tilde{t}_k^{(j)})$$

Generate spiking by integrate-and-fire.

Representing spike trains:

$$\underbrace{\Delta_i(t)}_{\text{spike train}} = \underbrace{\sum_{j=1}^{n_i}}_{\text{sum over spikes}} \underbrace{\delta(t - t_j^{(i)})}_{\text{"\delta-function" spike}}$$

What is a δ -Function:

- $\delta(t)$ is zero if $t \neq 0$ (infinitely sharp)

- Integrates to 1; $\int_{-\infty}^{\infty} dt \delta(t) = 1$

$\Rightarrow$ " $\delta(0)$ is infinite"

$\Rightarrow$ only precisely defined within integrals

- $\int_{-\infty}^{\infty} dt \delta(t) f(t) = f(0)$

- $\int_{-\infty}^{\infty} dt' \delta(t - t') f(t') = f(t) = \int_{-\infty}^{\infty} dt' \delta(t') f(t - t')$

This is convenient because

$$\begin{aligned} \sum_{j=1}^{n_i} K_s(t - t_j^{(i)}) &= \sum_{j=1}^{n_i} \int_{-\infty}^{\infty} dt' K_s(t - t_j^{(i)} - t') \delta(t') \\ &= \int_{-\infty}^{\infty} dt' \left(\sum_{j=1}^{n_i} \delta(t - t_j^{(i)} - t') \right) \cancel{\delta(t')} K_s(t') \\ &= \int_{-\infty}^{\infty} dt' \Delta_i(t - t') K_s(t') \\ &= (K_s * \Delta_i)(t) \end{aligned}$$

where the convolution of two functions is defined as

$$\begin{aligned} (f * g)(t) &= \int_{-\infty}^{\infty} dt' f(t - t') g(t') \\ &= \int_{-\infty}^{\infty} dt' f(t') g(t - t') = (g * f)(t) \end{aligned}$$

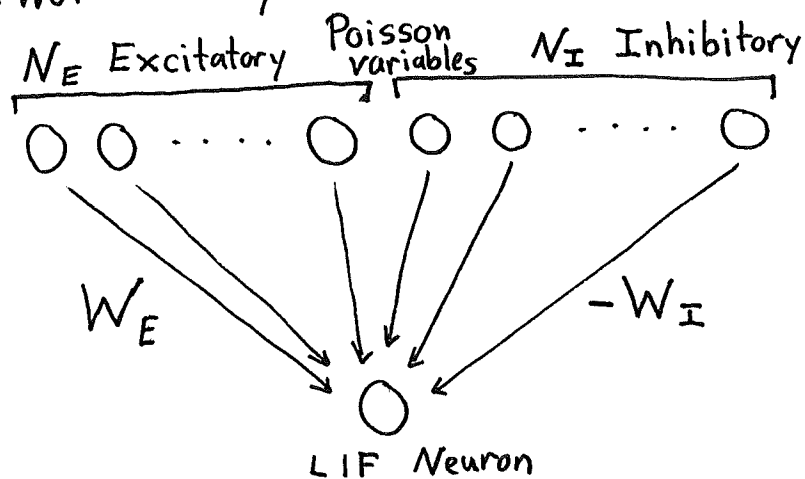
The coupled equations thereby become

$$C \frac{dV_i}{dt} = -G_L (V_i - E_L) + \sum_{j=1}^N W_{ij} (K_s * \Delta_j) \\ + \sum_{j=1}^M \mathcal{U}_{ij} (K_s * X_j)$$

where M is the number of input neurons, X_j is the spike train of the j^{th} input neuron, and the spike trains of the N recurrently connected neurons are computed from V_i by an integrate-and-fire mechanism.

Stochasticity in neural networks:

How do stochastic neural firing and neural network dynamics relate?



λ_E : Excitatory firing rate

λ_I : Inhibitory "rate"

If inputs have Poisson variability, how does output respond? Its input current is

$$I_i(t) = \sum_{j=1}^M \mathcal{U}_{ij} (X_j * K_s) = K_s * \left(\sum_{j=1}^M \mathcal{U}_{ij} X_j \right)$$

The temporal Kernel, K_s , smooths the Poisson variability.

We ignore it for now and set

$$K_s(t) = \delta(t) \Rightarrow I_i(t) = \sum_{j=1}^M \mathcal{U}_{ij} X_j(t)$$

As above, we assume excitatory (inhibitory) neurons have firing rate λ_E (λ_I) and weights W_E ($-W_I$).

The total current received in t to $t + \Delta t$ is

$$Q_{\Delta t, i}(t) = \int_t^{t+\Delta t} dt' I_i(t') = \sum_{j=1}^M \mathcal{U}_{ij} \int_t^{t+\Delta t} dt' X_j(t')$$

It's average over Poisson variability is

$$\begin{aligned} \langle Q_{\Delta t, i}(t) \rangle &= \sum_{j=1}^{N_E} W_E \lambda_E \Delta t - \sum_{j=N_E+1}^{N_E+N_I} W_I \lambda_I \Delta t \\ &= N_E W_E \lambda_E \Delta t - N_I W_I \lambda_I \Delta t \end{aligned}$$

The variances of independent random variables sum, $\text{Var}(aX+bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$, so

$$\begin{aligned}\text{Var}(Q_{\Delta t, i}(t)) &= \sum_{j=1}^M u_{ij}^2 \text{Var}\left(\int_t^{t+\Delta t} dt' X_j(t')\right) \\ &= \sum_{j=1}^{N_E} W_E^2 h_E \Delta t + \sum_{j=N_E+1}^{N_E+N_I} W_I^2 h_I \Delta t \\ &= N_E W_E^2 h_E \Delta t + N_I W_I^2 h_I \Delta t\end{aligned}$$

Limit of large networks:

For the mean to stay finite as N_E and N_I get large (go to infinity), we scale the weights,

$$W_E = \frac{\bar{W}_E}{N_E}, \quad W_I = \frac{\bar{W}_I}{N_I}$$

Then,

$$\begin{aligned}\langle Q_{\Delta t, i}(t) \rangle &= \bar{W}_E h_E \Delta t - \bar{W}_I h_I \Delta t \\ \text{Var}(Q_{\Delta t, i}(t)) &= \frac{1}{N_E} \bar{W}_E^2 h_E \Delta t + \frac{1}{N_I} \bar{W}_I^2 h_I \Delta t\end{aligned}$$

The variance goes to zero in an (infinitely) large system, and the Poisson variability has no effect. The downstream neuron acts like it would with $I_e = \bar{W}_E h_E - \bar{W}_I h_I$. It ~~either~~ either spikes regularly or is silent.

E/I balance:

The way out is to balance excitation and inhibition and achieve a zero mean.

$$\langle Q_{\Delta t, i}(t) \rangle = 0 \Rightarrow h_I = \frac{N_E W_E}{N_I W_I} h_E$$

Since the mean is already finite, we scale weights to maintain finite variance and trial-to-trial variability

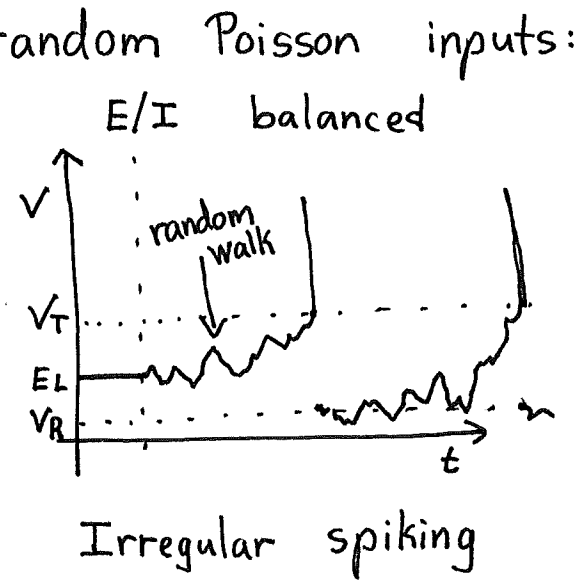
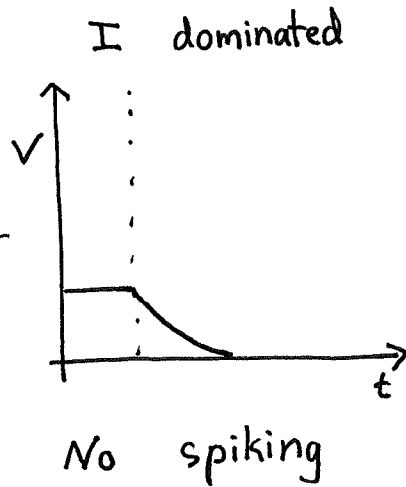
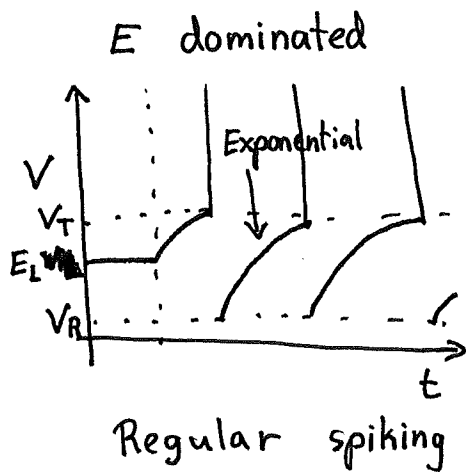
$$W_E = \frac{\bar{W}_E}{\sqrt{N_E}}, \quad W_I = \frac{\bar{W}_I}{\sqrt{N_I}}$$

In this balanced regime

$$\langle Q_{\Delta t, i}(t) \rangle = 0, \quad \text{Var}(Q_{\Delta t, i}(t)) = \bar{W}_E^2 \lambda_E \Delta t + \bar{W}_I^2 \lambda_I \Delta t$$

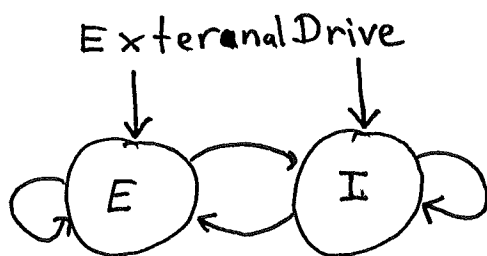
for large systems.

Response of LIF to many random Poisson inputs:



Dynamical balance:

Remarkably, van Vreeswijk and Sompolinsky showed that the balance condition is dynamically generated by recurrent neural networks



K : number of inputs

$$W = \frac{\bar{W}}{\sqrt{K}} : \text{weight scaling}$$

This is a candidate mechanism for the origin of Poisson-like variability.